

Laser Linewidth Measurement System

Thorlabs SA210 scanning Fabry-Pérot · SA201B controller · PicoScope 5242D
User & Hardware Manual

Application version: August 2026 (PySide6 / PyQtGraph interface)
Repository: github.com/alextishinin/linewidth-measurement · MIT License
Covers software commit 34b9bd4 and later

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1 System overview

This system measures the **linewidth (FWHM) of a continuous-wave laser** in real time. A Thorlabs SA210 scanning confocal Fabry-Pérot interferometer sweeps its cavity length with a piezo; every time the cavity passes through resonance with the laser, a transmission peak appears on the built-in photodiode. The SA201B controller generates the piezo ramp and amplifies the photodiode signal; a PicoScope 5242D digitizes it; and the Python application on this PC finds the peaks, calibrates the time axis into optical frequency, fits a Lorentzian to the main peak, and displays the linewidth live — together with its uncertainty, the cavity finesse, mode structure, and a running history. Every sweep is logged to CSV.

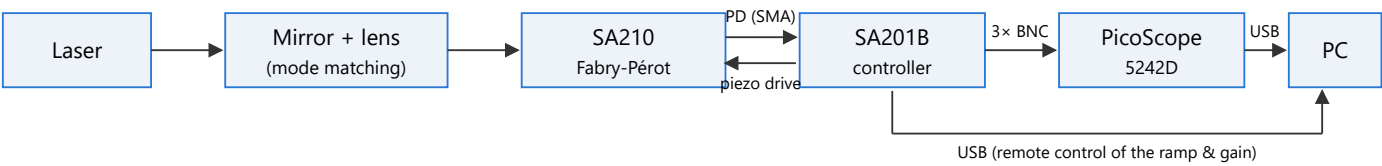


Figure 1 — signal chain.

Quantity	Value
Free spectral range (FSR)	10 GHz (SA210 series, confocal, $\text{FSR} = c/4d$)
Instrument resolution	~67 MHz (FSR / finesse, minimum spec finesse 150)
Measurable linewidth	≈67 MHz (instrument floor) up to a few GHz
Wavelength range (current cavity, SA210-8B)	820–1275 nm
Digitizer	15-bit, 2 MS/s default (16-bit in single-channel mode)
Update rate (live mode)	one analyzed sweep per ramp, typically 5–10 per second

Resolution floor. The measured width is the laser lineshape *convolved* with the ~67 MHz instrument function. A laser much narrower than 67 MHz reads ≈67 MHz regardless — that result means “narrower than the instrument can resolve.” The app flags this state as *instrument-limited* and, for broader lasers, also shows a deconvolved estimate (measured – 67 MHz).

2 Hardware

2.1 Parts list

Item	Role
Thorlabs SA210-8B scanning Fabry-Pérot (820–1275 nm)	the interferometer; photodiode is built into the head
Thorlabs SA201B controller	piezo ramp generator + photodiode transimpedance amplifier; USB-controlled
Pico Technology PicoScope 5242D	digitizer (any 5000D-series model works)
3× BNC cables + the SMA→BNC cable shipped with the SA210	signal wiring
Ø1" kinematic mount (Thorlabs KM100) for the SA210	tip/tilt alignment
Fold mirror (bench: Newport 10DER.2 broadband dielectric) + kinematic mount	beam steering into the cavity
f = 100 mm achromatic doublet, AR-coated 400–1100 nm	mode-matching lens (focus at the cavity center)
Windows 10/11 PC + monitor	runs this application (dedicated bench PC)

2.2 Wiring

From	To	Cable
SA210 permanently attached cable (piezo)	SA201B rear OUTPUT (0–45 V)	its own BNC
SA210 detector SMA connector	SA201B rear PD AMPLIFIER IN (labeled “photocurrent in” on some units — same input)	SMA→BNC (supplied)
SA201B front PD AMPLIFIER OUT	PicoScope Channel A	BNC
SA201B front TRIGGER OUT (5 V TTL at each ramp start)	PicoScope EXT trigger input	BNC
SA201B rear MONITOR OUT (ramp monitor)	PicoScope Channel B (<i>recommended, optional</i>)	BNC
SA201B USB-B and PicoScope USB-B	the PC	USB (scope on USB 3)

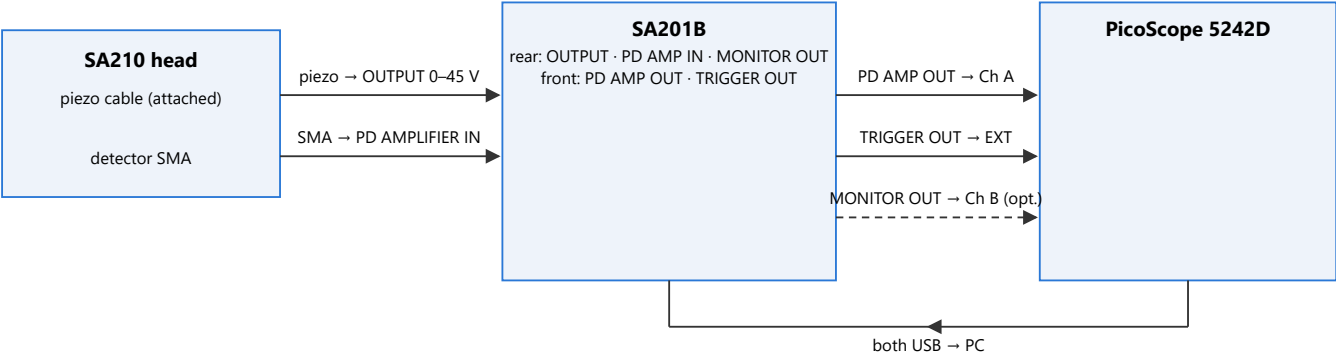


Figure 2 — cabling. The dashed Channel-B line is optional: with it, the app measures the true ramp duration every capture and resizes its window automatically; without it, launch with `--single-channel1` (the scope then digitizes at 16 bit instead of 15).

2.3 Optical layout

- Mount the SA210 in the Ø1" kinematic mount. Keep the input beam diameter around 1 mm.
- Steer the beam with the fold mirror into the SA210's input iris; place the $f = 100$ mm lens so its focus lands at the **cavity center, about 25 mm past the front flange**. Good mode matching minimizes transverse modes (section 4).
- Both irises on the SA210 help: close the input iris to center the beam, open the back iris while walking the mount.

Optical power. A few hundred μW to a few mW of CW power is plenty — the photodiode amplifier at its lowest gain (10k V/A) saturates around 5 V output well below damaging levels. If the app warns "*signal saturating*" at 10k gain, attenuate the beam. Consult the SA210 manual's damage threshold before sending in more than ~ 10 mW.

2.4 Power & bench notes

Three mains loads (SA201B, PC, monitor) run from one switched power strip mounted at the bench, so the whole station powers up together; the instruments enumerate over USB while the app's startup loop retries for up to a minute (section 3). The PC is dedicated to this instrument and launches the app automatically at logon.

3 Software installation & the kiosk setup

1. Windows 10/11 with Python 3.12+ (`winget install Python.Python.3.12`).
2. In the repository folder: `pip install -r requirements.txt` (installs numpy, scipy, PySide6, pyqtgraph, pyserial, picosdk).
3. Install **PicoSDK 64-bit** from picotech.com/downloads — it provides `ps5000a.dll` . The default install path `C:\Program Files\Pico Technology\SDK` is what `config.py` expects; edit `PICO_SDK_LIB` there if yours differs.
4. Wire the instruments per section 2.2, connect both USB cables, and run `run.bat` (or `python linewidth_live.py`). If your Python is installed elsewhere, edit the path inside `run.bat` .

Kiosk behavior on this bench PC

- **Auto-start:** a shortcut named *Linewidth Measurement* in the user's Startup folder (`shell:startup`) launches `run.bat` at every logon; the window opens maximized. Delete the shortcut to disable.
- **Desktop icon:** the same shortcut sits on the desktop for manual launch.
- **Startup retries:** after power-up, USB instruments can take a while to enumerate. The app retries opening the hardware 12 times at 5 s intervals (~1 minute) before giving up with a diagnostic message.
- **Only one owner:** close the PicoScope 7 desktop application before launching — only one program can hold the scope at a time.

4 Alignment procedure

1. Start the app (live mode, sawtooth running). Click **Align** (or press **t**): the controller switches to a **triangle** scan — the waveform Thorlabs recommends for alignment — and the big headline switches from linewidth to **peak height in volts**.
2. Center the beam: close the SA210 input iris, and steer the fold mirror until the transmitted spot is centered; open the iris again.
3. Watch the **Full sweep** graph while walking the SA210 mount's tip/tilt: make peaks appear, then maximize their height. Tall, sharp, *non-split* peaks indicate good coupling.
4. Now minimize the **transverse modes** percentage shown in the subtitle (and in the stats panel). Odd transverse modes of a confocal cavity appear exactly halfway between the main peaks; their relative height is a direct mode-matching meter: **<5% good**, 5–20% fair, >20% poor. Adjust tip/tilt and the lens focus position to push it down.
5. Click **Align** again to return to the sawtooth and normal measurement. Alignment sweeps are never logged and never enter the history plot.

Peak height and mode purity are both alignment criteria: height first (signal), then purity (single transverse mode). A perfectly mode-matched confocal cavity shows only the main comb.

5 Application tour — every control explained



Figure 3 — the application (dark theme, simulator data). Left: the three graphs, each with its own header bar and *reset view* button; the x-range control sits above the second graph. Right: the side panel with the headline readout, statistics, and all input controls.

5.1 The three graphs

Graph	Shows
Full sweep (top)	The photodiode amplifier output across one piezo ramp, in volts vs. milliseconds. Small triangles mark detected peaks. With a 30 V ramp expect the same pattern about 3 times — once per FSR. Use this panel while aligning.
Main peak, frequency calibrated (middle)	The same sweep re-plotted in optical frequency offset (MHz) around the analyzed peak, with the Lorentzian fit dashed on top. Several peaks inside one FSR = several longitudinal laser modes; their spacings are listed in the stats panel.
Linewidth (FWHM) history (bottom)	Measured FWHM vs. elapsed time. In live mode the last 3 minutes scroll; in single mode every <i>Run once</i> adds one point and old points stay.

Zooming (all three graphs, even while streaming)

- **Drag a box** with the left mouse button to zoom into it.
- **Scroll wheel** zooms around the cursor; dragging or scrolling *on an axis* rescales that axis alone.
- Any manual zoom **pauses auto-ranging for that graph** so the view stays put; its **reset view** button turns bold to show this. Zoomed-in views always show raw samples (the display only downsamples what a full view cannot resolve).
- **reset view** (per graph) or ☐ (all graphs) restores auto-ranging.

5.2 Graph-header controls

Control	Function
reset view (one per graph)	Restores that graph's automatic ranging after a manual zoom. Bold label = a manual zoom is active.
x-range dropdown (graph 2)	Chooses the frequency window of the calibrated-peak graph: · Auto — follows the fitted linewidth and any neighboring modes. · Full 10 GHz — the whole FSR (–5000...+5000 MHz), every mode of one order at once; useful for spotting half-FSR transverse modes. · Manual — the min/max boxes (MHz, each –5000...+5000, max > min) set the window; typing in either box switches to Manual automatically. The choice persists across sessions. A drag-zoom overrides it temporarily; reset view returns to the selected mode.

5.3 Side panel — buttons and input boxes

Control	Function
Light / Dark button (top right)	Toggles the color theme (the label names the theme you would switch <i>to</i>). Dark is the default; the choice is persisted. Also key d .
Full / Window button (top right)	Toggles full-screen — the window covers the whole monitor with no title bar or taskbar, suiting the dedicated bench display. Also F11 ; Esc exits. The state is persisted, so a kiosk left in full screen starts in full screen.
laser λ nm (100–5000)	The laser wavelength. Feeds the $\Delta\lambda$ display (the linewidth in pm/fm), the exported files, the CSV log, and the piezo calibration prior (section 7). Type a value and press Enter. Out-of-range entries are rejected with a message. Persisted.
ampl V (1–30)	Piezo ramp amplitude. Rule of thumb on the SA210: ~ 1 V per GHz of optical span, so 30 V $\approx$ 3 FSR. Applied to the SA201B over USB in the background (~ 0.5 s). Persisted.
offs V (0–15)	DC offset added to the ramp — slides the whole peak pattern along the sweep. Also nudged by the ← / → keys (± 0.25 V). Persisted.
sweep ms (10–100)	Ramp rise time at 1 $\times$ expansion, snapped to the SA201B's 0–200 step grid. Changing it automatically resizes the scope's capture window. Persisted.
expand dropdown	SA201B sweep expansion $\times 1$, $\times 2$, $\times 5$, $\times 10$, $\times 20$, $\times 50$, $\times 100$ — multiplies the sweep time for slow, fine scans. Long sweeps automatically coarsen the sample interval to keep captures manageable. Persisted.
PD gain (V/A) dropdown	Auto (default): the app steps the SA201B photodiode amplifier gain <i>down</i> when the 5 V output saturates and <i>up</i> when the tallest peak falls below ~ 0.35 V, with a 4 s cooldown between steps. 10k / 100k / 1M lock that transimpedance gain manually. The stats panel shows the gain actually in effect. Persisted.
Run once	In single mode (switching to it if needed): arms exactly one capture — the sweep is taken, analyzed, displayed, and acquisition stops again. Key r .
Mode: Live / Single	Toggles between continuous streaming and single-sweep mode (section 6). Key m .
Export data	Writes the currently displayed sweep and the linewidth history to CSV files in <code>exports\</code> (section 8.2). Key e .
Align: off / TRI	Toggles alignment mode — triangle scan, peak-height headline, nothing logged (section 4). Key t .

Every input box shows its valid range in its label. Values are applied optimistically — the UI reacts instantly while the controller write completes in the background; a failed write shows a warning, and the auto-reconnect re-applies your settings after a USB dropout.

5.4 Readouts and the stats panel

Headline (large number): the linewidth FWHM in MHz with its 1σ uncertainty, e.g. 66.5 ± 1.3 MHz . In live mode this is the median of the last 11 sweeps (spike-free); in single mode it is that sweep's own value. In Align mode it becomes the peak height in volts.

Subtitle: the same width converted to wavelength units at the current λ (e.g. $= 251$ fm @ 1064 nm), plus either *instrument-limited* (reading is at the ~ 67 MHz floor) or an estimated deconvolved laser width.

Warning area (red): analysis problems (*no signal...*), signal saturation, scope ADC clipping, controller-write failures, USB disconnects, and missing-trigger conditions appear here and clear automatically.

Stats block, line by line (lines appear only when relevant):

Line	Meaning
single sweep captured HH:MM:SS	timestamp of the last Run once
sweep-to-sweep scatter: $\pm x$ MHz	MAD-based scatter of recent raw sweeps — a direct measure of laser-cavity jitter (section 7)
half-max width	direct width at half maximum, without the Lorentzian fit — a fit-independent cross-check
fit R^2	goodness of the Lorentzian fit (1.000 = perfect)
effective finesse	$\text{FSR} \div \text{measured width}$; the SA210 spec is > 150 , so ≥ 150 confirms the instrument performs to spec
FSR spacing	measured peak-repeat period in ms and the resulting calibration slope in GHz/ms
modes in one FSR	number of longitudinal laser modes detected, with their spacings in MHz
transverse modes: N%	strongest half-FSR peak relative to the main peak — the alignment meter (good $< 5\%$ < fair $< 20\%$ < poor)
PD gain	the transimpedance gain currently in effect ([auto] if chosen automatically)
scan: ...	current ramp amplitude, sweep time $\times$ expansion, offset
sweep rate / window	ramps per second and the scope capture window
log: session_....csv	the CSV file this session is writing

5.5 Keyboard shortcuts

Key	Action	Key	Action
	Run one sweep (single mode)		Export data to CSV
	toggle Live / Single mode		snapshot (PNG + raw CSV)
	toggle Align mode		dark / light theme
	cycle PD gain manually		reset all zoomed views
	toggle auto-gain		pause / resume the display
 / 	DC offset ± 0.25 V		quit (identical to closing the window)
	toggle full screen		exit full screen

Keys are ignored while the cursor is in an input box.

6 Measurement modes

Live mode (default)

The scope captures every ramp; each triggered sweep is analyzed and displayed, the reported linewidth is the rolling median of the last 11 sweeps, and every sweep is appended to the session log. The history graph scrolls over the last 3 minutes (`--history-s` to change).

Single-sweep mode

Switch with **Mode** (`m`) or start with `--single`. The scope is *completely idle* — no data is acquired (the PicoScope LED stops flashing) until you press **Run once** (`r`), which arms exactly one capture (or N averaged sweeps with `--avg N`). The result freezes on screen; each run adds one point to the history, and old points never expire. Ideal for documenting discrete measurements.

Align mode

Overlay on either mode (`t`): triangle scan, peak-height headline, live transverse-mode meter, nothing logged. See section 4.

7 How the measurement works

7.1 From sweep to spectrum

The SA201B sweeps the cavity mirror with a sawtooth; the piezo moves $\lambda/4$ of mirror travel per free spectral range. The same laser line therefore re-appears every FSR, and the time between two repeats corresponds to exactly 10 GHz. The app measures that repeat period on every sweep and uses it to convert time to optical frequency — **the frequency axis is self-calibrating and assumes nothing about the ramp slope.**

7.2 The calibration guards

Finding the repeat period robustly is the heart of the instrument, because two effects can fool a naive search: the piezo's nonlinearity *chirps* the peak spacing across the sweep (spacings drift 10–20%), and a misaligned confocal cavity adds odd transverse modes **exactly halfway between the fundamentals** — a comb at FSR/2 that can look identical to the real one. The algorithm therefore:

1. builds candidate periods from tall-peak spacings and from the sweep's autocorrelation (each candidate also doubled, in case the raw candidate was the half-FSR comb);
2. rejects candidates that imply a finesse beyond $\sim 1.5\times$ the instrument spec — an impossibly *good* answer means the ruler is wrong — or that have no actual partner peak one period away;
3. picks among the survivors using the piezo's nominal response (~ 10 V per FSR at 1064 nm, scaled by the λ -box value since piezo travel per FSR is $\lambda/4$) as a soft prior;
4. measures the final spacing as the average of the left and right neighbor gaps of the analyzed peak, which cancels the chirp to first order;
5. falls back to the median period of the last 20 sweeps if a single sweep's calibration deviates $>4\%$ (flagged `cal-fallback` in the log).

The analyzed peak itself is chosen deterministically — a competitive interior peak nearest the sweep center, sticky to the previously tracked one — because with several near-equal peaks, “the tallest” would be a per-sweep noise lottery.

7.3 Lorentzian fit, jitter, and the median

A Lorentzian is fitted to the analyzed peak; its FWHM in calibrated units is the linewidth. A single peak transit lasts only $\sim \mu\text{s}$, so each sweep samples the instantaneous laser-cavity frequency jitter: raw per-sweep widths scatter by a few percent with occasional outliers. In live mode the app reports the **median of the last 11 sweeps** (~ 1.6 s of data; `--median N` to change, `--median 1` for raw), which is spike-free — while the raw value of every sweep is still logged unmodified. The observed scatter is itself displayed, as it directly measures the jitter.

7.4 The uncertainty budget

The headline's $\pm$ is a 1σ error combining, in quadrature:

- the Lorentzian fit's covariance on the width,
- the frequency-calibration error — the standard error of the FSR peak spacing over the last 20 sweeps,
- the standard error of the rolling median ($1.253 \cdot \sigma / \sqrt{N}$).

It is recorded in the log and in exported files.

8 Data: logging, export, snapshots, settings

Raw data is never altered. Statistics (the rolling median, the uncertainty) are always *added* in separate columns; the raw per-sweep measurement is logged unconditionally in every mode.

8.1 The session log (automatic)

Every launch creates `logs\session_YYYYMMDD_HHMMSS.csv` (disable with `--no-log`). One row per analyzed sweep; align-mode sweeps are excluded. The file is flushed every ~5 s and closed on exit. Columns:

Column	Content
<code>unix_time</code> , <code>iso_time</code>	capture timestamp (epoch seconds; ISO local time)
<code>linewidth_hz</code>	raw per-sweep fitted FWHM
<code>linewidth_median_hz</code>	the reported rolling median (live mode)
<code>linewidth_direct_hz</code>	direct half-maximum width, fit-independent
<code>deconvolved_hz</code>	measured – 67 MHz instrument function (≥ 0)
<code>finesse</code>	effective finesse ($\text{FSR} \div \text{width}$)
<code>fsr_period_s</code> , <code>hz_per_s</code>	measured FSR repeat period and calibration slope
<code>peak_v</code>	tallest peak height in volts
<code>n_modes</code>	longitudinal modes found in one FSR
<code>pd_gain_index</code>	0 / 1 / 2 = 10k / 100k / 1M V/A
<code>wavelength_nm</code> , <code>linewidth_pm</code>	λ -box value and the width as $\Delta\lambda$ in pm
<code>linewidth_err_hz</code>	the 1σ uncertainty (section 7.4)
<code>transverse_frac</code>	transverse-mode fraction 0–1
<code>flags</code>	ok, or any of <code>saturating</code> / <code>weak</code> / <code>cal-fallback</code> (semicolon-joined), or the analysis error message

8.2 Export (on demand)

Export data (`e`) writes to `exports\`:

- `sweep_YYYYMMDD_HHMMSS.csv` — the displayed sweep: columns `time_s`, `signal_v`, `freq_offset_mhz`, `fit_v`, preceded by a `#` metadata header (export time, λ , FSR, calibration slope, FWHM in MHz with 1σ , FWHM in wavelength units, finesse, transverse fraction, PD gain, mode).
- `history_YYYYMMDD_HHMMSS.csv` — the linewidth trend: `iso_time`, `elapsed_s`, `linewidth_mhz`, `linewidth_pm`.

Works in live or single mode; the file names appear in the status area after a successful export.

8.3 Snapshots

`s` saves to `logs\`: a **PNG screenshot** of the whole window (in the active theme) and a **raw CSV** of the displayed sweep (`time_s`, `signal_v`) — a quick lab-notebook record.

8.4 settings.json (automatic persistence)

On every exit the app writes `settings.json` next to the program and restores it on launch: wavelength, theme, ramp amplitude / offset / sweep time / expansion, gain choice (auto or fixed index), the graph-2 x-range mode and manual bounds, and the full-screen state. Precedence at startup: **explicit CLI flag** → **saved value** → **built-in default** — so a one-off `--wavelength-nm 780` wins for that session without overwriting your saved setup. Delete `settings.json` to return to factory defaults.

9 Measuring a different wavelength

9.1 Inside the current cavity's band (820–1275 nm): app only

Nothing physical needs to change. Type the new wavelength into the **laser λ nm** box and press Enter. This updates:

- the **$\Delta\lambda$ display** and all pm/fm conversions in exports and the log ($\Delta\lambda = \lambda^2 \cdot \Delta\nu / c$, so the same 67 MHz reads ~0.25 pm at 1064 nm but ~0.54 pm at 1550 nm);
- the **piezo calibration prior**: the mirror moves $\lambda/4$ per FSR, so the volts-per-FSR figure scales linearly with λ . The prior only breaks ties between calibration candidates — the frequency axis itself is always measured from the peak repeats, so even a wrong λ cannot corrupt the linewidth. It just weakens the disambiguation.

Then re-align (section 4) — steering and focus are wavelength-dependent.

9.2 Outside the band (e.g. 1550 nm): swap the cavity head

The current SA210-8B mirrors and photodiode cover 820–1275 nm. For 1550 nm, swap the interferometer head for an **SA210-12B** (1275–2000 nm). All SA210-series heads share the same 10 GHz FSR and the same connections, so:

1. **Physically**: swap the head in the same Ø1" mount; reconnect its piezo cable to SA201B **OUTPUT** and its detector SMA to **PD AMPLIFIER IN**. The photodiode is built into the head, so the detector swap is automatic — the SA201B amplifier doesn't care.
2. **Optics**: check every optic's coating for the new band. The bench's achromat is AR-coated 400–1100 nm — replace it with a 1050–1700 nm coated doublet of the same focal length; verify the fold mirror's coating covers 1550 nm (use an IR-band dielectric mirror, e.g. a 1280–1600 nm class coating, if not). An IR viewing card or camera is needed for alignment — the beam is invisible.
3. **In the app**: type 1550 in the λ box. Done — no code changes, no CLI flags; the FSR is unchanged so the calibration works as before, and the prior scales itself.
4. **Re-align** from scratch (section 4).

9.3 A different-FSR cavity (higher resolution)

Other Thorlabs scanning FP series (e.g. the SA200 family, FSR 1.5 GHz, resolution down to ~7.5 MHz) also work. Launch with the cavity's numbers, e.g.:

```
python linewidth_live.py --fsr-ghz 1.5 --instrument-mhz 7.5
```

Everything else — calibration, guards, display ranges, the “Full FSR” x-range label — follows those two flags automatically. Check the specific model's datasheet for its FSR, finesse, and wavelength band.

10 Command-line reference

All flags are optional; flags marked (*persisted*) default to your last-used in-app value. `run.bat` passes any arguments through.

Flag	Meaning
<code>--single</code>	start in single-sweep mode
<code>--port COM3</code>	SA201B COM port (default: autodetect by USB ID)
<code>--no-controller</code>	don't talk to the SA201B; set the ramp on its touchscreen (pass <code>--rise-ms</code> if not 10 ms)
<code>--single-channel</code>	MONITOR OUT not wired; photodiode only, 16-bit
<code>--amplitude 30</code>	ramp amplitude V, 1–30 (<i>persisted</i>)
<code>--offset 0</code>	ramp DC offset V, 0–15 (<i>persisted</i>)
<code>--risetime-step 0</code>	sweep-time step 0–200 → 10–100 ms (<i>persisted as sweep ms</i>)
<code>--sweep-expand 0</code>	expansion index 0–6 = $1 \times 100 \times$ (<i>persisted</i>)
<code>--pdgain auto</code>	<code>auto</code> , or fixed index <code>0 / 1 / 2</code> = 10k/100k/1M V/A (<i>persisted</i>)
<code>--rise-ms 10</code>	sweep time when running with <code>--no-controller</code>
<code>--dt-us 0.5</code>	target sample interval ($0.5 \mu\text{s} \approx 45$ samples across a 67 MHz feature)
<code>--window-ms</code>	fixed capture window (default: auto from the sweep time)
<code>--avg 1</code>	average N triggered sweeps before analysis
<code>--median 11</code>	live-mode rolling-median length (1 = report raw sweeps)
<code>--wavelength-nm 1064</code>	laser wavelength (<i>persisted</i>)
<code>--theme dark</code>	<code>dark</code> or <code>light</code> (<i>persisted</i>)
<code>--fsr-ghz 10</code>	cavity FSR (SA210 = 10)
<code>--instrument-mhz 67</code>	instrument resolution (SA210 spec = 67)
<code>--history-s 180</code>	live-mode history window, seconds
<code>--no-log</code>	disable the session CSV log

11 Troubleshooting

Symptom	Cause & fix
<i>no trigger — is SA201B TRIGGER OUT connected...</i>	The scope's EXT input sees no ramp-start pulse. Check the TRIGGER OUT → EXT BNC and that the ramp is actually running (SA201B screen).
<i>no signal — check laser alignment into the SA210</i>	Triggered sweeps but no peaks above ~0.35 V at any gain: the beam misses the cavity, the laser is off/blocked, or the power is far too low. Re-align (section 4).
<i>! signal saturating</i>	PD amplifier output at the 5 V rail: auto-gain will step down; at 10k V/A already, attenuate the beam.
<i>! scope ADC clipped</i>	The digitizer range clipped a sample; usually accompanies saturation and clears when the level drops.
<i>! SA201B USB disconnected — retrying...</i>	The controller's serial link dropped. Measurement continues (the ramp free-runs); the app reconnects automatically and re-applies amplitude, offset, sweep time, expansion, gain and waveform. Check the USB cable if it persists.
Startup fails with <code>PICO_NOT_FOUND</code>	Another program owns the scope (close PicoScope 7), the USB cable dropped — or a previous process was force-killed mid-capture , which wedges the driver with a stuck request. In that last case only a reboot frees the unit. Prevention: always close the app with its window's X or <code>q</code> ; never end it from Task Manager while it is capturing. The app prints <code>[exit] scope released</code> on every clean shutdown.
App doesn't appear right after power-on	USB enumeration can lag a cold boot; the launcher retries for ~1 min. If it gave up, double-click the desktop icon.
Peaks halfway between the main peaks	Odd transverse modes — an alignment/mode-matching indicator, not extra laser modes. Minimize the <i>transverse modes</i> % (section 4). They are automatically excluded from the mode count and calibration.
More “modes” than expected, all equal height	Verify the frequency axis before believing it: check that the peak pattern repeats ~once per 10 V of ramp (the SA210's nominal piezo response at 1 μm). Two laser modes spaced near FSR/2 (mod FSR) can mimic a doubled comb — perturb the laser (e.g. pump power): real two-mode beat spacing moves, the piezo's repeat spacing does not.
Linewidth reads ≈67 MHz and won't go lower	That's the instrument floor — the laser is narrower than the SA210 can resolve (the subtitle says <i>instrument-limited</i>). The reading is an upper bound.
History shows occasional single-sweep spikes in the log	Normal: raw sweeps sample μs-scale jitter (a few % scatter, ~3% of sweeps spike). The reported/displayed median is spike-free; both are in the log (section 8.1).

12 Maintenance, tests & specifications

Housekeeping

- **Logs grow:** live mode writes several rows per second (~a few MB per hour). Old `logs\session_*.csv` files can be archived or deleted freely; the app never re-reads them.
- **Reset the configuration:** delete `settings.json` (next to the program) to return every persisted control to its default.
- **Update the software:** `git pull` in `C:\linewidthmeasurement`. Every push to the repository runs the analysis self-test in CI on Python 3.11–3.13.
- **Self-test:** `python test_analysis.py` verifies the whole measurement pipeline against a synthetic laser — no hardware needed.

File map

File	Role
<code>linewidth_live.py</code>	the application (PySide6 + PyQtGraph UI, acquisition thread)
<code>analysis.py</code>	peak finding, FSR self-calibration, Lorentzian fit, uncertainty
<code>sa201b.py</code> / <code>pico5000a.py</code>	SA201B serial driver / PicoScope block-mode driver
<code>capture.py</code>	one-sweep data container
<code>simulator.py</code> + <code>test_analysis.py</code>	synthetic source + self-test (used by CI)
<code>config.py</code>	FSR, resolution, piezo prior, ports, themes
<code>run.bat</code> , <code>icon.ico</code>	launcher (Startup + desktop shortcuts point here)
<code>logs\</code> , <code>exports\</code> , <code>settings.json</code>	session logs & snapshots, exported CSVs, persisted settings (not in git)

Key specifications

Parameter	Value
FSR / resolution / finesse (SA210)	10 GHz / ~67 MHz / > 150
Cavity band (current head, SA210-8B)	820–1275 nm
Piezo response (nominal, soft prior only)	~10 V per FSR at 1064 nm, scales with λ
Ramp	1–30 V, 0–15 V offset, 10–100 ms $\times$ 1–100 expansion, sawtooth (triangle in Align)
Trigger	SA201B TRIGGER OUT, 5 V TTL, rising edge at 1.5 V on EXT
Sampling	0.5 μ s default (2 MS/s); 15-bit two-channel, 16-bit single-channel
Capture window	sweep time $\times$ 1.25 + 2 ms, auto-tracked via MONITOR OUT
PD amplifier gains	10k / 100k / 1M V/A (SA201B), auto-managed by default
Reported value (live)	median of last 11 sweeps, 1σ uncertainty from fit + calibration + median SE

SA210, SA201B are Thorlabs products; PicoScope is a Pico Technology product — consult their manuals for device-level specifications. This manual documents the measurement application (MIT license) at github.com/alextishinin/linewidth-measurement.